INCH-POUND

MIL-DTL-24640/23D
22 August 2002
SUPERSEDING
MIL-C-24640/23C
26 June 1995

DETAIL SPECIFICATION SHEET

CABLE, ELECTRIC, 600 VOLTS, LIGHT-WEIGHT, MULTI-CONDUCTOR, AWG 18, TYPE MXCW (INCLUDING VARIATION TYPE MXCOW)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24640 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

Construction (Watertight)

First - Copper conductor, uncoated AWG 18, 7 strands, in accordance with ASTM B 8,

Second - Mica glass tape insulation 0.0029 ±.0002 inch (50 percent overlap maximum and minimum overlap of 45 percent). Quality conformance testing shall be in accordance with MIL-W-81044, prior to cabling.

Third - Opaque-pigmented polyimide tape .001 inch (50 percent overlap minimum).

Fourth - Opaque-pigmented polyimide tape .001 inch with 0.0001 inch Fluorinated Ethylene Propylene [FEP] (50 percent overlap minimum).

Fifth - Modified aromatic polyimide resin coating 0.0005 inch minimum (.001 inch nominal). Identification codes applied using Method 1 or 3.

Sixth - The required number of component wires shall be cabled together. Cabling sequence shall be consecutive, starting with number 1 from center outward. Fillers may be used as necessary, to form a round cable.

Seventh - Binder tape applied helically with overlap.

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CONSTRUCTION VARIANT FOR TYPE MXCW

Eighth - Crosslinked polyolefin jacket. (See Table I for thickness).

CONSTRUCTION VARIANT FOR TYPE MXCOW

Eighth - Braided shield, tin coated copper strands. Braid must be constructed to meet Surface Transfer Impedance requirements as listed herein.
Ninth - Binder tape applied helically with overlap.
Tenth - Cross linked polyolefin jacket. (See Table II for thickness).

TABLE I. Construction details.

PIN M24640/23	Type number	Number of conductors	Jacket thickness (inch) (min)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)
-01UN	MXCW-7	7	0.038	0.295	0.319	90.0
-02UN	MXCW-10	10	0.041	0.375	0.405	139.0
-03UN	MXCW-14	14	0.041	0.402	0.434	173.0
-04UN	MXCW-19	19	0.041	0.440	0.474	215.0
-05UN	MXCW-24	24	0.044	0.520	0.560	295.0
-06UN	MXCW-30	30	0.044	0.547	0.589	340.0
-07UN	MXCW-37	37	0.044	0.584	0.630	398.0
-08UN	MXCW-44	44	0.044	0.656	0.708	492.0
-09UN	MXCW-61	61	0.047	0.729	0.785	640.0

TABLE II. Construction details - overall shield.

PIN M24640/23	Type number	Number of conductors	Diameter over shield (inch) (nom)	Jacket thickness (inch) (min)	Cable diameter (inch) (min)	(max)	Cable weight (pounds per 1000 feet) (max)
-01UO	MXCOW-7	7	0.226	0.038	0.340	0.366	140.0
-02UO	MXCOW-10	10	0.297	0.041	0.415	0.447	185.0
-03UO	MXCOW-14	14	0.325	0.041	0.440	0.474	225.0
-04UO	MXCOW-19	19	0.370	0.041	0.477	0.515	280.0
-05UO	MXCOW-24	24	0.441	0.044	0.557	0.601	362.0
-06UO	MXCOW-30	30	0.469	0.044	0.584	0.630	399.0
-07UO	MXCOW-37	37	0.514	0.044	0.622	0.670	466.0
-08UO	MXCOW-44	44	0.585	0.047	0.697	0.751	574.0
-09UO	MXCOW-61	61	0.652	0.047	0.757	0.817	715.0

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REQUIREMENTS:

Qualification required.
Construction details: See Tables I and II

Finished wire: For polyimide tape constructions, the following additional tests shall be applicable in accordance with MIL-W-81381:

Temperature rating: 150°C (302°F) maximum conductor temperature.
Voltage rating: 600 volts (rms) at sea level.
Blocking: Oven temperature, 200 ± 2°C (392 ± 3.6°F) no sticking of conductors.
Flammability: 3 seconds (max) after flame.
3.0 inches (max) flame travel.
No flaming of tissue paper.
Humidity resistance: 5 megohms for 1000 feet, minimum insulation resistance after humidity exposure.
Impulse dielectric test: 100 percent test. Impulse voltage at 8.0 kV.
Insulation resistance: 2500 megohms for 1000 feet (min).
Polyimide cure test: No cracking.
Shrinkage: 0.062 inch maximum at 180°C
Surface resistance: 5 megohms inches (min), initial and final readings.
Thermal shock: Oven temperature, 150 ± 2°C (302 ± 3.6°F).
Maximum change in measurement: 0.031 inch.
Wet dielectric test: 2500 volts (rms).

EXAMINATION AND TESTS:

Basic electrical:	Requirements:
Conductor Resistance - ohms/1000 feet at 25°C, maximum	7.07
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2000
Conductor to shield (MXCOW only)	1000
Shield to shield	Not applicable
Insulation resistance - megohms-1000 feet, minimum	
Conductor to conductor	2500
Conductor to overall shield (MXCOW only)	2500
Shield to shield (component shield only)	Not applicable
Shield to shield (component shield to overall shield	Not applicable
Conductor and shield continuity (type MXCOW only)	No failure
Jacket flaws (type MXCOW only)	No failure

Group A:
Visual and dimensional.....No failure
Capacitance:
Mutual capacitance - at 1 MHz,

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picofarads/foot (pF/ft), maximum.....Not applicable
Capacitance unbalance - percent, maximum .....Not applicable
Characteristic impedance - at 1 MHz, ohms.....Not applicable
Watertightness (see Table IX of MIL-DTL-24640 for limits of water leakage).....No failure
Weight .....No failure

Group B:
Cold bending.....No failure
Drip .....No failure
Cross link proof test (percent maximum)
Jacket 50
Tear - pounds per inch thickness, minimum.....35
Physicals (unaged)
Jacket
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160
Attenuation - at 3 MHz, dB/100 feet, maximum .....Not applicable
Gas flame (1 hour) .....No failure

Group C:
Physicals (aged)
Jacket
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged, minimum 60
Permanence of printing (conductor) - method 1 only cycles, minimum 25
Permanence of printing (jacket) - cycles, minimum.....125
Shield - conformance to material, construction and coverage (type MXCOW only).....No failure
Heat distortion - percent of unaged, maximum 30
Cable sealant removability.....No failure
Shrinkage .....No failure
Mutual inductance - dB/1000 feet, maximum .....Not applicable
Surface transfer impedance (type MXCOW only)
Milliohms per meter, maximum 700
EMP response, dB, minimum 60
Pulse response time - ms/500 feet
Rise time, maximum .....Not applicable
Decay time, maximum .....Not applicable

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

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Aging and compatibility (cable)	No failure
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Hot oil immersion	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
Fillers	45
Insulation	10
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure
Fungus Resistance	No failure
Abrasion resistance	No failure
Gas Flame (3 hour)	Not applicable
Component wire	No failure

Unit ordering length: 7 - 19 500 feet nominal.
24 - 61 250 feet nominal

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2307-023)

Review Activities:
Army - AR, AV, CR, CR4
Navy - CG, EC
DLA - IS

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